



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam, Trimester: Fall 2025

Course Code: CSE-3521, Course Title: Database Management Systems

Total Marks: 40, Duration: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

- 1 a) Consider the relation $R(A, B, C, D, E, F, G)$ with functional dependencies:
- $A \rightarrow BC$
 - $B \rightarrow DE$
 - $CD \rightarrow F$
 - $F \rightarrow G$
 - $EG \rightarrow A$
- i) Using Armstrong's axioms only, **determine** whether $AB \rightarrow CD$ can be inferred. 2
 - ii) **Find** the closure of BD and AC . 2
- b) Consider the relation $R(A, B, C, D, E, F, G, H)$ with the functional dependencies:
- $AB \rightarrow C$
 - $AC \rightarrow D$
 - $D \rightarrow E$
 - $E \rightarrow A$
 - $BF \rightarrow G$
 - $CG \rightarrow H$
 - $H \rightarrow B$
- i) **Find** out all possible candidate keys. 5
 - ii) **Identify** prime attributes and non-prime attributes. 1
 - iii) **Determine** the maximum normalized form (upto BCNF) of relation R . 2
- c) Given the set of functional dependencies of relation $R(A, B, C, D, E, F, G)$: 4
- $ABC \rightarrow DE$
 - $B \rightarrow CF$
 - $D \rightarrow EG$
 - $AF \rightarrow C$
 - $C \rightarrow B$
 - $E \rightarrow A$
- Compute** the minimal cover of the set of functional dependencies. 2
- d) A relation $R(A, B, C, D, E)$ with functional dependencies: 2
- $A \rightarrow BC$
 - $C \rightarrow D$
 - $D \rightarrow E$
- is decomposed into $R_1(A, B, C)$, $R_2(C, D)$, and $R_3(D, E)$. Find whether the decomposition is **lossless** or not.

e) Relation **R(A, B, C, D, E, F, G, H, I)** with the following dependencies **2**

$A \rightarrow BC$

$C \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$DE \rightarrow I$

$I \rightarrow A$

$AF \rightarrow D$

$CI \rightarrow G$

is decomposed into the following **three relations**:

$R_1(A, B, C, F)$

$R_2(C, D, E, I)$

$R_3(F, G, B)$

Determine whether the given decomposition is dependency preserving or not.

- 2 a) Consider a database table STUDENTS with the following attributes:
STUDENTS(StudentID, Name, Dept, Semester, CGPA, AdmissionYear, DOB)
The table is physically stored on disk in ascending order of StudentID.
However, different users frequently need to retrieve records sorted by attributes other than StudentID. The following access requirements are common:
- University management retrieves student lists sorted by CGPA.
 - Department offices retrieve student records sorted by Name.
 - The examination office generates reports sorted by AdmissionYear.
- i) **Explain** why changing the physical order of the table for each sorting requirement is not practical. **2**
- ii) **Propose** a solution that will meet all the requirements on the same table. **Describe** how your proposed solution resolves the issues. **2**
- b) Consider the following data coming from left to right. Construct a B+ Tree of order 3 for the incoming data and show all necessary steps while inserting. **6**
50, 18, 72, 6, 29, 83, 41, 95, 14, 67
- 3 a) **Explain** the ACID properties of transactions. For each property, describe what could go wrong if it is violated. **6**
- b) What are the 3 classic problems of concurrency in transaction. Explain any 2 with proper example. **4**